COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 / Data Structures and Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	July 15, 2026	**DATE SUBMITTED:**	July 15, 2026
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SEATWORK 2.1 **Memory Diagram Exercise**

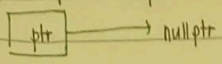
1. PROCEDURE OUTPUT

1. **Draw the memory diagram for the following:**

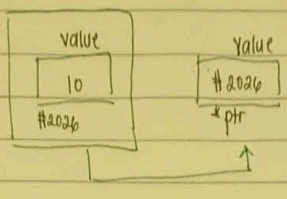
```
1. int *ptr = nullptr;
2. int value = 10;
   int  *ptr= &value;
3. int *ptr = new int;
4. int *ptr = new int(10);
```

SW 2.1: Memory Diagram Exercises

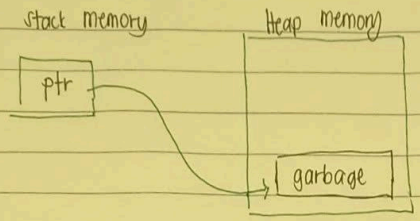
1.) int *ptr = nullptr;



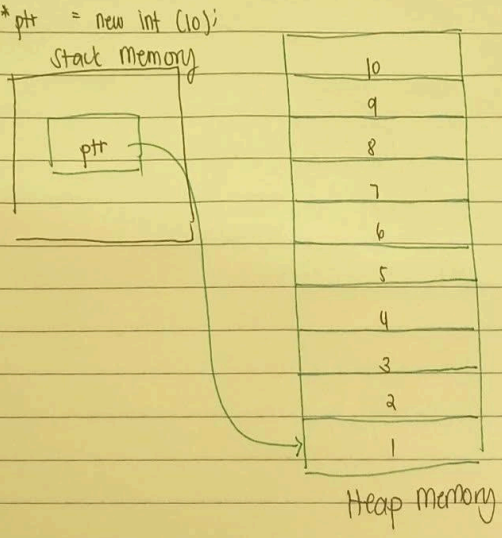
2.) int value = 10;
int *ptr = &value;



3.) int *ptr = new int;



4.) int *ptr = new int (10);



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2. **Differentiate Constructor, Destructor, Copy Constructor, Copy Assignment Operator.**

- **Constructor =** initializes an object when it is created.
- **Destructor =** cleans up resources before an object is destroyed.
- **Copy Constructor =** creates new object and copies the existing code
- **Copy Assignment Operator =** copies the content of the existing object into another existing object.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

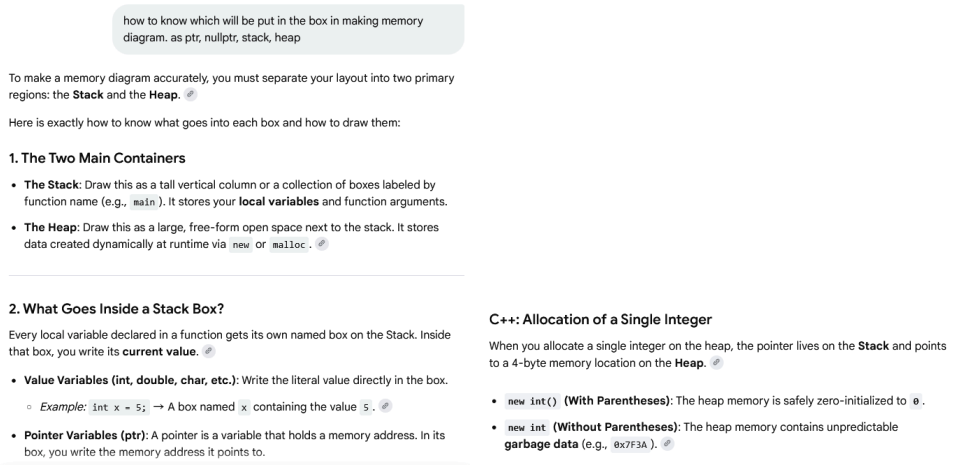
Tools Used: **Google Overview**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **For asking concepts and clarifying questions I have in mind.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:



how to know which will be put in the box in making memory diagram. as ptr, nullptr, stack, heap

To make a memory diagram accurately, you must separate your layout into two primary regions: the **Stack** and the **Heap**.

Here is exactly how to know what goes into each box and how to draw them:

1. The Two Main Containers

- **The Stack**: Draw this as a tall vertical column or a collection of boxes labeled by function name (e.g., `main`). It stores your **local variables** and function arguments.
- **The Heap**: Draw this as a large, free-form open space next to the stack. It stores data created dynamically at runtime via `new` or `malloc`.

2. What Goes Inside a Stack Box?

Every local variable declared in a function gets its own named box on the Stack. Inside that box, you write its **current value**.

- **Value Variables (int, double, char, etc.)**: Write the literal value directly in the box.
  - *Example:* `int x = 5;` → A box named `x` containing the value `5`.
- **Pointer Variables (ptr)**: A pointer is a variable that holds a memory address. In its box, you write the memory address it points to.

C++: Allocation of a Single Integer

When you allocate a single integer on the heap, the pointer lives on the **Stack** and points to a 4-byte memory location on the **Heap**.

- `new int()` **(With Parentheses)**: The heap memory is safely zero-initialized to `0`.
- `new int` **(Without Parentheses)**: The heap memory contains unpredictable **garbage data** (e.g., `0x7F3A`).

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.



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